

JEE ADVANCED CUMULATIVE TEST SERIES

44
YEARS
OF EXCELLENCE



NARAYANA
IIT-JEE ACADEMY-INDIA

Sec: JR.IIT_*CO-SC(MODEL-A)

Time: 3 Hrs

Date: 14-07-24

Max. Marks: 180

14-07-24_JR.IIT_STAR CO-SC(MODEL-A)_JEE ADV_CAT-8_SYLLABUS

PHYSICS:

100% DESIGNATED TOPICS: Quadratic equations

CHEMISTRY:

100% DESIGNATED TOPICS: Thermometry, Basic maths, Vectors,
Motion in a straight line

MATHEMATICS:

100% DESIGNATED TOPICS: Gaseous state, Thermochemistry

THE NARAYANA GROUP

PHYSICS

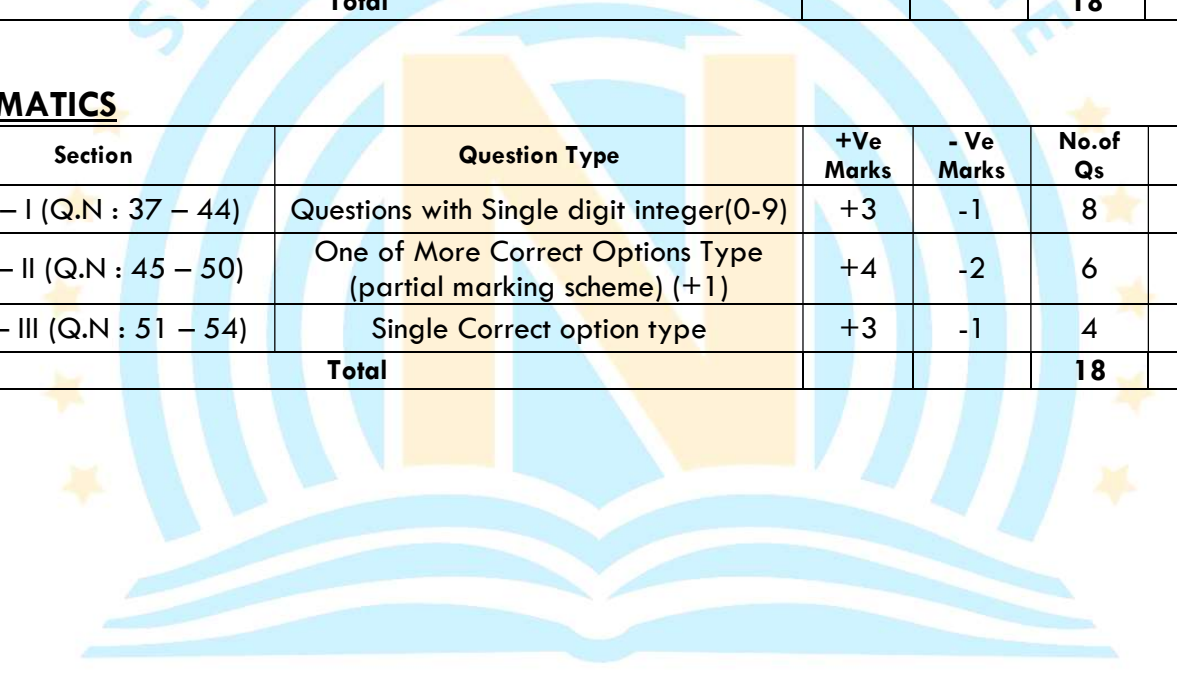
Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 8)	Questions with Single digit integer(0-9)	+3	-1	8	24
Sec – II (Q.N : 9 – 14)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 15 – 18)	Single Correct option type	+3	-1	4	12
Total				18	60

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 – 26)	Questions with Single digit integer(0-9)	+3	-1	8	24
Sec – II (Q.N : 27 – 32)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 33 – 36)	Single Correct option type	+3	-1	4	12
Total				18	60

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 – 44)	Questions with Single digit integer(0-9)	+3	-1	8	24
Sec – II (Q.N : 45 – 50)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 51 – 54)	Single Correct option type	+3	-1	4	12
Total				18	60



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MATHEMATICS**MAX.MARKS: 60****SECTION – 1 (Maximum Marks: 24)**

This section contains SIX (08) questions.

The answer to each question is **A SINGLE DIGIT INTEGER** ranging from **0 TO 9**, **BOTH INCLUSIVE**.

For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric

keypad in the place designated to enter answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If only the correct option is chosen.

Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)

Negative Marks: -1 In all other cases.

- Let $f(x) = x^2 + ax + b$. If $\forall x \in R$, there exist a real value of y such that $f(y) = f(x) + y$, then find the maximum value of $10a$.
- Find sum of all possible values of the real parameter 'b' if the difference between the largest and smallest values of the function $f(x) = x^2 - 2bx + 1$ in the segment $[0, 1]$ is 4
- If a, b, c are the roots of the cubic $x^3 - x^2 + 2x - 3 = 0$, then find the absolute value of the expression $\frac{a}{a+bc} + \frac{b}{b+ac} + \frac{c}{c+ab}$
- Let $f(x) = kx^2 + x(3-4k) - 12$. If the set of values of k for which $f(x) < 0 \forall x \in (-3, 3)$ and $f(-4) > 0$ is $(p, q]$ then find the value of $(4p + 3q)$
- Let $f(x) = \sqrt{\frac{x^2 + ax + 4}{x^2 + bx + 16}}$ is defined for all real x and if the number of possible ordered pairs (a, b) (where $a, b \in I$) are K then find sum of digits of K .
- Find the number of integral values of k for which the equation $x^2 - 4|x| + 3 - |k-1| = 0$ has four distinct real roots.
- The number of real solution of equation $2021x^6 + 2021x^5 - 2020x^4 + x^3 + 2020x^2 + 2021x - 2021 = 0$
- Let $x_1 < x_2 < x_3$ be the three real roots of the equation $\sqrt{2014}x^3 - 4029x^2 + 2 = 0$, then $x_2(x_1 + x_3)$ is

SECTION - 2 (Maximum Marks : 24)

This section contains SIX (06) questions.

Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).

For each question, choose the correct option(s) to answer the question.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks: -2 In all other cases.

9. If the equation $\sin^2 \theta - a \sin \theta + b = 0$ has only one solution in $(0, \pi)$, then

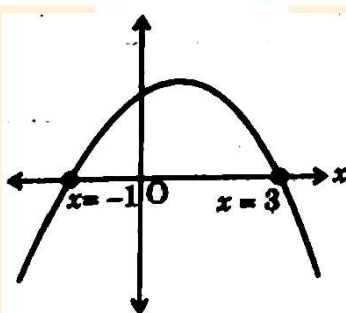
A) $a \in (-\infty, 1] \cup [2, \infty)$

B) $b \in (-\infty, 0] \cup [1, \infty)$

C) $a = 1 + b$

D) $a + b = 1$

10. Consider the graph of quadratic polynomial $y = ax^2 + bx + c$ as shown below.



Which of the following is(are) correct?

A) $\frac{a-b+c}{abc} = 0$

B) $abc(9a+3b+c) < 0$

C) $\frac{a+3b+9c}{abc} < 0$

D) $ab(a-3b+9c) > 0$

11. $P(x)$ is a cubic polynomial such that $P(1) = 1, P(2) = 2, P(3) = 3$ and $P(4) = 16$, then:

A) $P(0) = -12$

B) $P(5) = 53$

C) $P(0) = 12$

D) $P(-1) = -49$

12. If all the three equations $x^2 + ax + 10 = 0$, $x^2 + bx + 8 = 0$ and $x^2 + (2a + 3b)x + 60 = 0$, where $a, b \in \mathbb{R}$, have a common root, then value of $(a - b)$ can be

A) -1

B) 0

C) 1

D) 2

13. If all values of x which satisfies the inequality $\log_{1/3}(x^2 + 2px + p^2 + 1) \geq 0$ also satisfy the inequality $kx^2 + kx - k^2 \leq 0$ for all real values of k , then all possible values of p lies in the interval

A) $[-1, 1]$

B) $[0, 1]$

C) $[0, 2]$

D) $[-2, 0]$

14. Let the roots of the equation $x^3 + ax + a = 0, a \in \mathbb{R} - \{0\}$ be α, β and γ such that

$$\frac{\alpha^2}{\beta} + \frac{\beta^2}{\gamma} + \frac{\gamma^2}{\alpha} + 8 = 0 \text{ and } \alpha < \beta < \gamma, \text{ then}$$

- A) $\alpha = -1 - \sqrt{5}$ B) $\beta = 1 - \sqrt{5}$ C) $\gamma = 1 + \sqrt{5}$ D) $a = -8$

SECTION - 3 (Maximum Marks : 12)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

15. If α, β, γ are the roots of the cubic $x^3 - 2x + 3 = 0$, then the value of

$$\frac{1}{\alpha^3 + \beta^3 + 6} + \frac{1}{\beta^3 + \gamma^3 + 6} + \frac{1}{\gamma^3 + \alpha^3 + 6} \text{ equals}$$

- A) $\frac{1}{3}$ B) $-\frac{1}{3}$ C) $\frac{1}{2}$ D) $-\frac{1}{2}$

16. Let $P(x) = x^2 + \frac{4x}{3} + \log_{10}(4.9)$, $A = \prod_{i=1}^{12} P(a_i)$ where $a_1, a_2, \dots, a_{12}$ are positive reals and

$B = \prod_{j=1}^{13} P(b_j)$ where $b_1, b_2, \dots, b_{13}$ are non-positive reals, then which one of the following is always correct?

- A) $A > 0, B > 0$ B) $A > 0, B < 0$ C) $A < 0, B > 0$ D) $A < 0, B < 0$

17. A monic quadratic trinomial $P(x)$ is such that $P(x) = 0$ and $P(P(P(x))) = 0$ have a common root, then:

- A) $P(0).P(1) > 0$ B) $P(0).P(1) < 0$ C) $P(0).P(1) = 0$ D) None of these

18. Let $p, q, r, s \in \mathbb{R}$ (the set of all real numbers) then the number of real roots of the equation

$$(x^2 + px - 3q)(x^2 - rx + q)(x^2 - sx + 2q) = 0 \text{ are:}$$

- A) Exactly two B) Atleast two C) Atmost four D) Exactly four

PHYSICS

MAX.MARKS: 60

SECTION – 1 (Maximum Marks: 24)

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The answer to each question is **A SINGLE DIGIT INTEGER** ranging from **0 TO 9**, **BOTH INCLUSIVE**.

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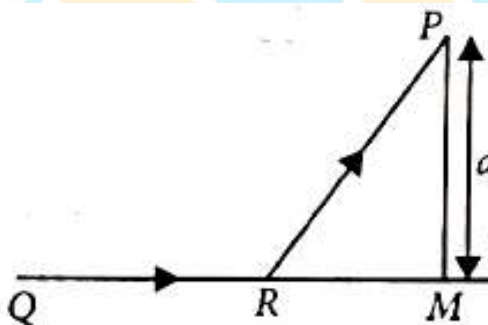
Answer to each question will be evaluated according to the following marking scheme:

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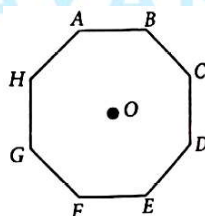
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)

Negative Marks: -1 In all other cases.

19. Two trains 'A' and 'B' of length ' l ' and ' $4l$ ' are travelling into a tunnel of length ' L ' in parallel tracks from opposite directions with velocities 108km/h and 72km/h, respectively. If train 'A' takes 35s less time than train 'B' to cross the tunnel then, length ' L ' of tunnel is (Given $L = 60l$) x km then find value of $5x$
20. A man in a car at location Q on a straight highway is moving with speed v . He decides to reach a point P in a field at a distance d equal to $\sqrt{3}$ km from the highway (point M) as shown in the figure. Speed of the car in the field is half to that on the highway. What should be the distance RM (in km), so that the time taken to reach P is minimum?



21. An object moving with a speed of 6.25ms^{-1} , is decelerated at a rate given by $\frac{dv}{dt} = -2.5\sqrt{v}$, where v is the instantaneous speed. The time taken by the object, to come to rest (in sec), would be
22. In an octagon ABCDEFGH of equal side, if $\vec{AO} = 2\hat{i} + 3\hat{j} - 4\hat{k}$ then the sum of $\vec{AB} + \vec{AC} + \vec{AD} + \vec{AE} + \vec{AF} + \vec{AG} + \vec{AH}$ is equal to $x\hat{i} + y\hat{j} + z\hat{k}$ find $2x + z$ is equal to



23. For a train engine moving with speed of 20ms^{-1} , the driver must apply brakes at a distance of 500 m before the station for the train to come to rest at the station. If the brakes were applied at half of this distance, the train engine would cross the station with speed $10\sqrt{x}\text{ms}^{-1}$. The value of x is _____ (Assuming same retardation is produced by brakes)

24. Three particles P, Q and R are moving along the vectors $\vec{A} = \hat{i} + \hat{j}, \vec{B} = \hat{j} + \hat{k}$ and $\vec{C} = -\hat{i} + \hat{j}$ respectively. They strike on a point and start to move in different directions. Now, particle P is moving normal to the plane which contains vector $\vec{A}$ and $\vec{B}$. Similarly particle Q is moving normal to the plane which contains vector $\vec{A}$ and $\vec{C}$. The angle between the direction of motion of P and Q is $\cos^{-1}\left(\frac{1}{\sqrt{x}}\right)$. Then the value of x is _____
25. The sum of two forces $\vec{P}$ and $\vec{Q}$ is $\vec{R}$ such that $|\vec{R}| = |\vec{P}|$. The angle θ (in radian) that the resultant of $2\vec{P}$ and $\vec{Q}$ will make with $\vec{Q}$ is $\frac{\pi}{n}$. Find $n = ?$
26. A ball is dropped from rest at height $4h$. After it has fallen a distance d , a second ball is dropped from rest at height h . What should be the ratio $\frac{d}{h}$ so that the balls hit the ground at the same time

SECTION - 2 (Maximum Marks : 24)

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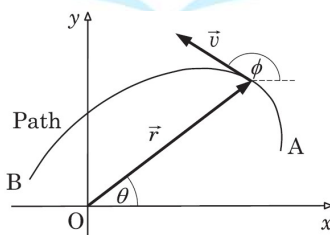
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

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Negative Marks: -2 In all other cases.

27. At a particular instant of time, position vector $\vec{r}$, velocity vector $\vec{v}$ and angular position θ of a particle traversing a path AB are shown in the figure. Here ϕ is the angle made by the velocity vector with the positive x -axis. Which of the following statements is/are correct?



A) Modulus of angular velocity is $\frac{d\theta}{dt} = \frac{v \sin(\phi - \theta)}{r}$.

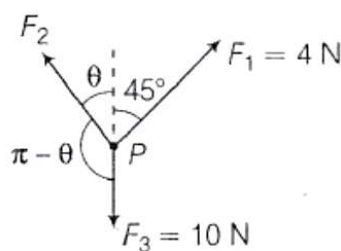
B) Modulus of tangential component of acceleration is $r \frac{d^2\theta}{dt^2}$

C) Modulus of normal component of acceleration is $v \frac{d\theta}{dt}$

D) Modulus of normal component of acceleration is $v \frac{d\phi}{dt}$

28. A point moves rectilinearly with deceleration whose modulus depends on the velocity v of the particle as $\omega = a\sqrt{v}$, where a is a positive constant. At the initial moment the velocity of the point is equal to v_0 . Then
- A) The distance traversed before it stops is $\left(\frac{2}{3a}\right)v_0^{3/2}$
- B) The distance traversed before it stops is $\left(\frac{1}{3}a\right)v_0^{3/2}$
- C) The time taken to stop is $\frac{2\sqrt{v_0}}{a}$
- D) The time taken to stop is $\sqrt{v_0}/a$
29. A particle moving continuously in the positive x-direction passes the positions $x = 9$ m and 17 m at the instants $t = 1$ s and 3 s respectively. Its average velocities in the time intervals [1 s, 3 s] and [0 s, 6 s] are equal. Which of the following statements is/are correct?
- A) It was at $x = 5$ m at $t = 0$ s
- B) It is moving with a uniform speed
- C) Average velocity in the interval [3 s, 6 s] is 4 m/s.
- D) Information is insufficient to decide.
30. A particle, starting from rest is first accelerated for time t_1 with constant acceleration a_1 and then stops in time t_2 with constant retardation a_2 . Let v_1 be the average velocity in this case and s_1 the total displacement. Now, the same particle, starting again from rest is accelerated for the same time t_1 with constant acceleration $2a_1$ and finally comes to rest with constant retardation a_2 in time t_3 . If v_2 is the average velocity in this case and s_2 the total displacement. Then
- A) $s_2 = 2s_1$ B) $2s_1 < s_2 < 4s_1$ C) $v_2 = 2v_1$ D) $2v_1 < v_1 < 4v_1$
31. Two particles P and Q move in straight line AB towards each other. P starts from A with velocity u_1 and an acceleration a_1 . Q starts from B with velocity u_2 and acceleration a_2 . They pass each other at the midpoint of AB and arrive at the other ends of AB with equal speeds.
- A) They meet at midpoint at time $t = \frac{2(u_2 - u_1)}{(a_1 - a_2)}$
- B) The length of path specified i.e. AB is $l = \frac{4(u_2 - u_1)(a_1u_2 - a_2u_1)}{(a_1 - a_2)^2}$
- C) They reach the other ends of AB with equal speeds if $(u_2 + u_1)(a_1 - a_2) = 8(a_1u_2 - a_2u_1)$
- D) They reach the other ends of AB with equal speeds if $(u_2 - u_1)(a_1 + a_2) = 8(a_2u_1 - a_1u_2)$

32. Three forces F_1, F_2 and F_3 are acting on particle P (shown in figure). If particle P is in equilibrium, then



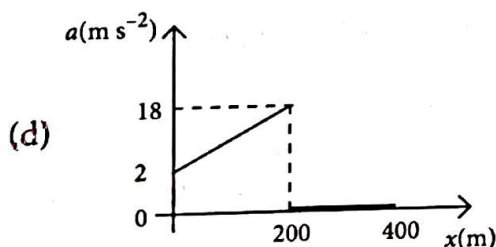
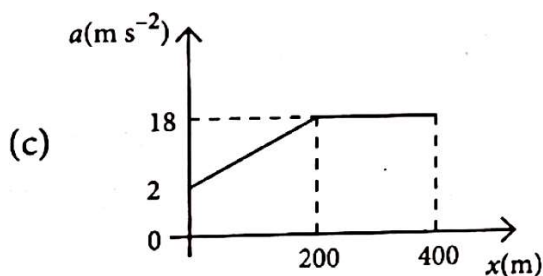
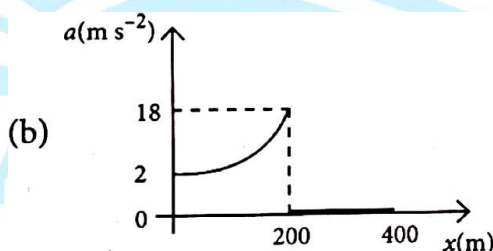
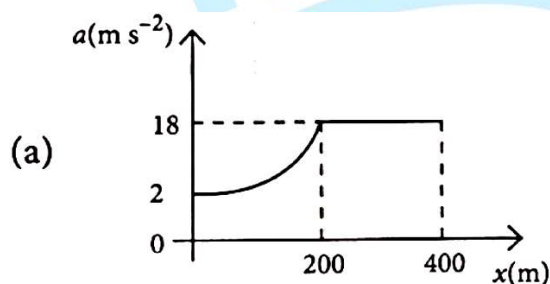
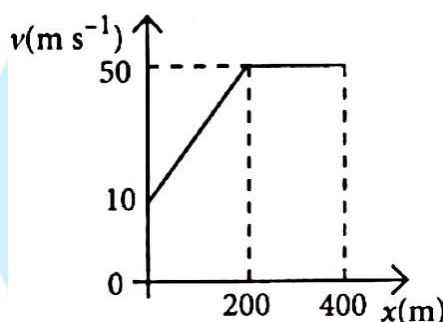
- A) $\tan \theta = \frac{2}{(5\sqrt{2} - 2)}$ B) $F_2 \sin \theta = 2\sqrt{2}N$ C) $F_2 \cos \theta = 2\sqrt{2}N$ D) $\tan \theta = \left(\frac{5\sqrt{2} - 2}{2} \right)$

SECTION - 3 (Maximum Marks : 12)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

33. The velocity – displacement graph describing the motion of a bicycle is shown in the figure. The acceleration – displacement graph of the bicycle's motion is best described by



34. Two vectors $\vec{P}$ and $\vec{Q}$ have equal magnitudes. If the magnitude of $\vec{P} + \vec{Q}$ is n times the magnitude of $\vec{P} - \vec{Q}$, then angle between $\vec{P}$ and $\vec{Q}$ is

- A) $\cos^{-1}\left(\frac{n-1}{n+1}\right)$ B) $\sin^{-1}\left(\frac{n-1}{n+1}\right)$ C) $\cos^{-1}\left(\frac{n^2-1}{n^2+1}\right)$ D) $\sin^{-1}\left(\frac{n^2-1}{n^2+1}\right)$

35. **Statement I:** Two forces $(\vec{P} + \vec{Q})$ and $(\vec{P} - \vec{Q})$ where $\vec{P} \perp \vec{Q}$, when act at angle θ_1 to each other, the magnitude of their resultant is $\sqrt{3(P^2 + Q^2)}$, when they act at an angle θ_2 , the magnitude of their resultant becomes $\sqrt{2(P^2 + Q^2)}$. This is possible only when $\theta_1 < \theta_2$.

Statement II: In the situation given above, $\theta_1 = 60^\circ$ and $\theta_2 = 90^\circ$.

In the light of the above statements, choose the most appropriate answer from the options given below

- A) Statement I is false but statement II is true.
 B) Both Statement I and Statement II are false.
 C) Both Statement I and Statement II are true.
 D) Statement I is true but Statement II is false.

36. Let $|\vec{A}_1| = 3$, $|\vec{A}_2| = 5$ and $|\vec{A}_1 + \vec{A}_2| = 5$. The value of $(2\vec{A}_1 + 3\vec{A}_2) \cdot (3\vec{A}_1 - 2\vec{A}_2)$ is

- A) -106.5 B) -99.5 C) -118.5 D) -112.5

CHEMISTRY**MAX.MARKS: 60****SECTION – 1 (Maximum Marks: 24)**

This section contains SIX (08) questions.

The answer to each question is **A SINGLE DIGIT INTEGER** ranging from **0 TO 9**, **BOTH INCLUSIVE**.

For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric

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Negative Marks: -1 In all other cases.

37. How many of the following statement(s) are correct?

(I) The compression factor, Z , approaches 1 at low pressures, but does so with different slopes. For a perfect gas, the slope is zero, but real gases may have either positive or negative slopes, and the slope may vary with temperature.

(II) critical temperature of O_2 is less than CO_2

(III) When Z vs P is plotted, methane shows less deviation than ethane

(IV) The critical temperature of oxygen is 155K. it signifies that it is impossible to produce liquid oxygen by compression alone if the temperature is greater than 155 K.

(V) Although a and b are not precisely defined molecular properties, they correlate with physical properties such as critical temperature, vapor pressure, and enthalpy of vaporization that reflect the strength of intermolecular interactions.

(VI) for a Van der Waals gas, Perfect gas isotherms are obtained at high temperatures and large molar volumes.

38. How many of the following are correct statement(s)/option(s)?

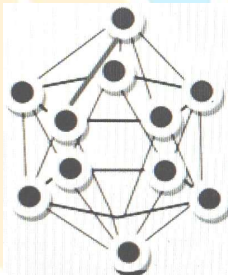
(I) The standard enthalpies of formation of elements in their reference states are zero at all temperatures

(II) if the enthalpy of formation of $HBr(aq)$ is found to be -122 kJ mol^{-1} , then the formation of $Br^-(aq)$, $\Delta H_f(Br^-, aq) = -122 \text{ kJ mol}^{-1}$.

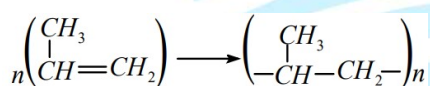
(III) If Lattice enthalpy and enthalpy of solution of $NaCl$ are 788 kJ mol^{-1} and 4 kJ mol^{-1} , respectively, then the hydration enthalpy of $NaCl$ is -784 kJ mol^{-1}

(IV) All exothermic reactions have negative free energy change

39. 2 moles of Ne gas and 5 moles of He gas, both samples having average velocity $7 \times 10^2 \text{ m/s}$, are mixed. Find the average translational kinetic energy per mole of the given gas mixture (in joules). Report your answer as 'X' (rounded off to nearest integer) where,
- $$X = \frac{\text{(Average translational KE in joules)}}{1237.5}$$
40. The diffusion coefficient of an ideal gas is proportional to its mean free path and mean speed. The absolute temperature of an ideal gas is increased 4 times and its pressure is increased 4 times. As a result, the diffusion coefficient of this gas increases x times. The value of x is:
41. One litre of a gas at 300 atm and 473 K is compressed to a pressure of 600 atm and 344 K. The compressibility factors are found to be 1.250 and 1.357 respectively at the initial and final states. The final volume of the real gas is 'x' L. Then 10x value is (round off to nearest integer)?
42. Some element, "X", exists in different allotropic forms. If, one of its gaseous allotropic forms contain icosahedral units (icosahedral is a regular shape with 12 corners and 20 faces) with X atoms at all 12 corners and all bonds being equivalent. If "A" is the heat evolved at constant pressure (in kJ/mol) of X atom producing such an allotropic form and if the bond dissociation enthalpy of X-X bond is 300 kJ/mol, then report A/375?



43. The polymerization of propene to linear polypropene is represented by the reaction



Where n has large integral value, the average enthalpies of bond dissociation for (C=C) and (C-C) at 298 K are +590 and +331 kJ mol^{-1} , respectively. The enthalpy of polymerization is -360 kJ mol^{-1} . Find the value of n.

44. The enthalpy of formation of $\text{C}_2\text{H}_5\text{OH}$ is -66 kcal/mol and the enthalpy of combustion of CH_3OCH_3 (g) is -348 kcal/mol . Enthalpy of formation of H_2O and CO_2 are -68 kcal/mol and -94 kcal/mol respectively. Calculate ΔH for isomerization of $\text{C}_2\text{H}_5\text{OH(l)}$ into CH_3OCH_3 (g) in kcal/mol. Report your answer as sum of digits till you get the single digit answer.

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Negative Marks: -2 In all other cases.

45. A space capsule is filled with neon gas at 1.00 atm and 290 K. The gas effuses through a pin-hole into outer space at such a rate that the pressure drops by 0.3 torr/sec

a) If the capsule were filled with ammonia at the same temperature and pressure, what would be the rate of pressure drop?

b) If the capsule were filled with 30.0 mol% helium, 20.0 mol% oxygen and 50.0 mol% nitrogen at a total pressure of 1.00 atm and a temperature of 290 K, what would be the corresponding rate of pressure drop?

If the answers are X and Y respectively, then choose X and/or Y from the choices given below?

A) 0.32 torr/sec (B) 0.29torr/sec (C) 0.96torr/sec (D) 1.4torr/sec

46. A gas at 250 K and 15 atm has a molar volume 12 per cent smaller than that calculated from the perfect gas law. Choose the correct statement(s)?

(A) the compression factor under these conditions 0.12

(B) the molar volume of the gas is $1.2\text{dm}^3\text{ mol}^{-1}$

(C) repulsive forces are dominating in the sample

(D) attractive forces are dominant

47. A volume of 100 ml of 0.5N H_2SO_4 solution is neutralized with 200 ml of 0.2M NH_4OH in a constant pressure calorimeter which resulted in $1.4^\circ C$ rise in temperature. The heat capacity of the calorimeter system is $1.5 kJ / ^\circ C$. Some useful thermochemical equations are:



Which of the following statements are correct?

- A) Enthalpy of neutralization of HCl vs. NH_4OH is $-52.5 kJ / mol$.
- B) Enthalpy of dissociation (ionization) of NH_4OH is $4.5 kJ / mol$
- C) Enthalpy of dissociation (ionization) of CH_3COOH is $4.6 kJ / mol$
- D) ΔH for $2H_2O(l) \rightarrow 2H^+(aq) + 2OH^-(aq)$ is 114 kJ
48. The enthalpy change for a reaction depends upon:
- A) physical state of the reactants and products
- B) use of different reactants for the same products
- C) nature of intermediate reaction steps
- D) internal energies of reactants and products
49. Choose the correct statement(s) with reference to the following two experiments.
- (Expt. 1):** When 100 mL of 1.0 M HCl was mixed with 100 mL of 1.0 M $NaOH$ in an insulated beaker at constant pressure, a temperature increase of $5.7^\circ C$ was measured for the beaker and its contents. Because the enthalpy of neutralization of a strong acid with a strong base is a constant ($-57 kJ/mol$), this experiment could be used to measure the calorimeter constant.
- (Expt. 2):** In a second experiment, 100 mL of 2.0 M acetic acid ($K_a = 2.0 \times 10^{-5}$) was mixed with 100 mL of 1.0 M $NaOH$ (under identical conditions to Expt. 1) where a temperature rise of $5.6^\circ C$ was measured. (Consider heat capacity of all solutions as $4.2 J g^{-1} K^{-1}$ and density of all solutions as $1.0 g mL^{-1}$, $\log 2 = 0.3$)
- A) calorimeter constant is $16 J / ^\circ C$
- B) Enthalpy of dissociation of acetic acid is $10 kJ/mol$
- C) The pH of the solution after Expt. 2 is 4.7
- D) The pH of the acetic acid solution before Expt. 2 is 2.2

50. Two flask A and B have equal volumes. Flask A contains hydrogen at 600 K while flask B has same mass of CH_4 at 300 K. Then choose the correct options
- (A) In flask A the molecules move faster than B
 (B) In flask B the molecules move faster than A
 (C) Flask A contains greater number of molecules than B
 (D) Flask B contains more molecules than A

SECTION - 3 (Maximum Marks : 12)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

51. The standard enthalpy of formation of the metallocene bis(benzene)chromium was measured in a calorimeter. It was found for the reaction $\text{Cr}(\text{C}_6\text{H}_6)_2(\text{s}) \rightarrow \text{Cr}(\text{s}) + 2 \text{C}_6\text{H}_6(\text{g})$ that $\Delta U_r(583 \text{ K}) = +8.0 \text{ kJ mol}^{-1}$. Find the standard enthalpy of formation of the compound at 583 K? Given, the constant-pressure molar heat capacity of benzene is $136.1 \text{ J K}^{-1} \text{ mol}^{-1}$ in its liquid range and $81.67 \text{ J K}^{-1} \text{ mol}^{-1}$ as a gas.
- (a) +116 kJ/mol (b) -116 kJ/mol (C) +58 kJ/mol (d) -58 kJ/mol
52. $2\text{X}_{(\text{g})} + 3\text{Y}_{(\text{g})} \rightarrow \text{Z}_{(\text{g})}$ X and Y are insoluble and inert to water and Z forms basic solution. In an experiment 3 mole each of 'X' and 'Y' are allowed to react in 15 L flask at 500 K. When reaction is complete. 5L of water is added to the flask and temperature is reduced to 300 K. The pressure in the flask in atm is (neglect aqueous tension)
- (A) 1.64 (B) 2.46 (C) 4.92 (D) 3.28
53. For a certain gas which deviates a little from ideal behaviour, the values of density, ρ were measured at different values of pressure, P. The plot of $\frac{P}{\rho}$ on the Y-axis versus P on the X-axis was nonlinear and had an intercept on the Y-axis, which was equal to (where M = Molar mass)
- (A) $\frac{RT}{M}$ (B) $\frac{M}{RT}$ (C) RT (D) $\frac{RT}{V}$
54. If volume of CO_2 gas is negligible, then the pressure exerted by 1 mol of CO_2 gas at 300K is ($a = 3.0 \text{ atm.L}^2.\text{mol}^{-2}$; $R = 0.08 \text{ L.atm.mol}^{-1}\text{K}^{-1}$)
- (A) 84.4 atm (B) 24.4 atm (C) 12.2 atm (D) 48.0 atm